PAT 521 Advanced Psychoacoustics
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Transitivity and Timbre Similarity Judgment

Abstract

Timbre researchers have often asked participants to judge the similarities of audio stimuli in order to construct a multidimensional timbre space. An assumption built into the timbre space approach is that participants' timbre similarity judgments are self-consistent and satisfy the requirements of a metric: non-negativity, symmetry, and the triangle inequality. Any timbre similarity judgments that display intransitivity (A>B, B>C, C>A) violate the triangle inequality. The present pilot study attempts to establish the transitivity or intransitivity of similarity judgments of complex timbral stimuli. Participants are presented with a series of forced-choice questions in which they must select which of two complex test tones is more similar to a reference tone. The stimulus sets are constructed so that each test stimulus is more similar to the reference than another test stimulus along two auditory dimensions and less similar along a third dimension. There is therefore no ground truth similarity for any stimulus set. The patterns of answers are analyzed for self-consistency across trials, transitivity, and correlation with the three varied auditory attributes. The patterns of answers observed in this pilot study do not support the intransitivity of timbre similarity judgments. The observed answers also do not support the expected distribution for reliable transitivity. A wide range of self-consistency was observed among participants, and no single auditory attribute dominates the similarity judgments. Further development of the pilot is needed to make any claims about the transitivity of timbre similarity judgments.

1. Introduction

While perceived audio attributes like loudness, pitch, and duration have measurable physical correlates, there is not just one physical quantity that can be varied to induce a change in the perception of timbre. Timbre perception seems tied to a large and loose collection of auditory attributes. Different individuals weight each of these attributes differently when perceiving timbre. Defining timbre has been the subject of sixty years of research.

To better understand timbre perception, researchers often ask participants to judge the similarities of audio stimuli in order to construct a multidimensional *timbre space*. A timbre space treats sound stimuli as points in a geometric space (see Figure 1); the judged similarity of the stimuli is treated as a distance between the stimuli in the space. The timbre space approach is motivated by the desire to find measurable audio attributes that covary with each axis of the timbre space. An assumption built into the timbre space approach is that participants' timbre similarity judgments are self-consistent and

satisfy the mathematical requirements of a metric. There are three requirements for a metric, or distance:

1) Distances are always positive.
2) The distance from x to y is the same as the distance from y to x.
3) If x, y, and z are three points in space, the distance from x to z is always less than or equal to the sum of the distances from x to y and from y to z (the triangle inequality).

Any timbre similarity judgments that display *asymmetry* (sound x is very similar to y, but y is not very similar to x) violate the second requirement for a metric. Any timbre similarity judgments that display *intransitivity* (sound y is more similar to x than z, and sound z is more similar to x than y) violate the third requirement for a metric. Studying the transitivity of timbre similarity judgments therefore offers one way to explore the validity of the timbre space approach.

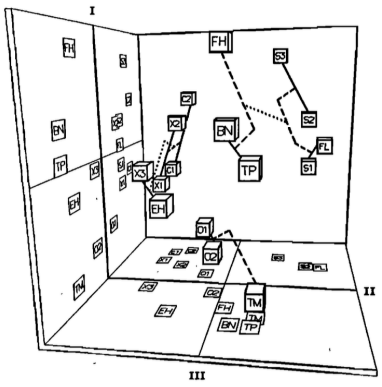


Figure 1: A timbre space with each stimulus represented as a cube, from (Grey, 1977).

While many psychoacoustic studies embrace the timbre space approach, and while some have demonstrated the self-consistency and context-independence of timbre similarity judgments (Grey, 1977; McAdams, Winsberg, Donnadieu, De Soete, & Krimphoff, 1995; McAdams, 1999), other studies have found that human similarity judgments of complex stimuli display intransitivity, inconsistency, asymmetry, and other context effects (Grey, 1977; Goldstone, Medin, & Halberstadt, 1997).

I borrowed an experimental method from psychologists Goldstone, Medin, and Halberstadt (Goldstone, Medin, & Halberstadt, 1997) and conducted a pilot study to test the transitivity of timbre similarity judgments using complex auditory stimuli. I designed timbral stimuli that vary along three auditory dimensions shown by prior research to be relevant for timbre perception. I then asked participants a series of two-way forced-choice questions to ascertain whether their timbre similarity judgments display transitivity.

In the following sections, I describe **2.** related work, **3.** the experimental method, **4** results and analysis, and **5** future work to be done.

2. Related Work

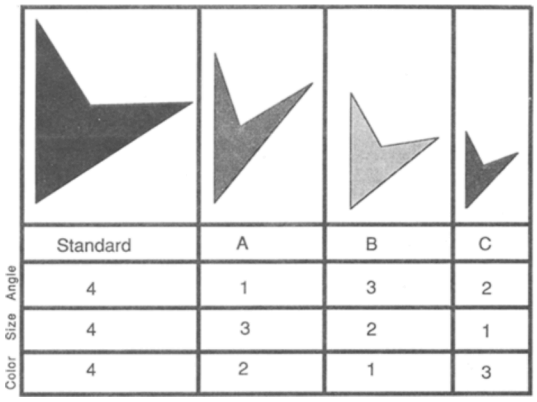
In this pilot study, I primarily draw on the timbre research of (Grey, 1977), (McAdams, Winsberg, Donnadieu, De Soete, & Krimphoff, 1995; McAdams, 1999), and (Peeters, Giordano, Susini, Misdariis, & McAdams, 2011) and the psychology research of (Goldstone, Medin, & Halberstadt, 1997) and (Hepting, Bin Amer, & Yao, 2019).

In 1977, Grey introduced the timbre space approach (Grey, 1977). He gathered forced-choice similarity judgments on pairs of synthesized tones, averaged the judgments across participants, and performed multidimensional scaling and clustering to map of the tones in a three-dimensional timbre space (see Figure 1). Grey found that multiple sessions with the same participant yielded the same judgments but that experimental context affected judgments. Grey's 1977 experiment forms the basis of much ensuing timbre similarity research.

Stephen McAdams and his team at IRCAM have conducted research on timbre similarity for over thirty years (McAdams, Winsberg, Donnadieu, De Soete, & Krimphoff, 1995; McAdams, 1999). McAdams' work in the 1990s adopted Grey's 1977 methodology, using synthesized instrument tones, multidimensional scaling, and clustering to convert participants' pairwise similarity judgments into a timbre space. The team identified correlates for the axes of timbre space (log attack time, spectral centroid, spectral smoothness, and spectral flux) while delineating five distinct listener groups that weighted each of these attributes differently. The team found that timbre similarity judgments were robust to context change. Their approach continued to assume the metric nature of similarity judgments and continues to influence timbre research.

In 2011, a team led by Peeters released the Timbre Toolbox, a Matlab library that extracts audio features relevant for timbre research (Peeters, Giordano, Susini, Misdariis, & McAdams, 2011). The development of the Timbre Toolbox included a meta-analysis of over fifty auditory attributes, resulting in a method for constructing small, orthogonal sets of audio features for use in timbre research. I followed this method in selecting the three auditory attributes I varied in designing the stimuli for this experiment.

Psychologists Goldstone et al. challenged assumptions about the self-consistency and transitivity of similarity judgments of complex visual stimuli (Goldstone, Medin, & Halberstadt, 1997). In one experiment, Goldstone et al. found that human similarity judgment of complex visual stimuli violated transitivity. A person presented with three complex visual stimuli and a reference would judge B to be more similar to the reference than A, C more similar than B, and A more similar than C (see Figure 2), a violation of the third requirement for a metric. Participants seemed to shift the dimensions to which they paid the most attention within a single trial, with no one strategy dominating. Because of their findings, the team argued against the idea of multidimensional spaces as representations of complex perceptual phenomena. I borrowed the methodology of Goldstone et al.'s Experiment 1 in conducting the current pilot study.



	Standard	A	B	C
Angle	4	1	3	2
Size	4	3	2	1
Color	4	2	1	3

Figure 2: Complex visual stimuli used by (Goldstone, Medin, & Halberstadt, 1997) to test the transitivity of similarity judgments.

In 2019, psychologists Hepting et al. conducted an experiment studying how participants grouped similar complex visual stimuli (Hepting, Bin Amer, & Yao, 2019). The team asked participants to sort 356 photographs into piles of similar photographs. The team was unable to discern a common strategy by which these similarity judgments were made across participants, reflecting the ambiguity and complexity of the task. This might indicate that we can expect auditory stimulus comparison to be equally complex.

These studies offer conflicting accounts of the metric and transitive nature of human similarity judgments. The current pilot study attempts to apply Goldstone's methodology, originally using visual stimuli, to complex timbral stimuli in order to explore how transitive human similarity judgments of timbre might be.

3. Experiment

The method for this pilot study is largely borrowed from (Goldstone, Medin, & Halberstadt, 1997), Experiment 1, but with the visual stimuli replaced by audio stimuli. The experiment attempts to establish the transitivity or intransitivity of participants' similarity judgments of timbral stimuli.

3.1 Hypotheses

Hypothesis:

> Timbre similarity judgments display systematic intransitivities and violate weak stochastic transitivity. The distribution of transitive and intransitive answers to forced-choice similarity questions with complex timbral stimuli will be significantly similar to the distribution of answers found by Goldstone et al. for complex visual stimuli.

Null hypothesis:

> Timbre similarity judgments do not display systematic intransitivities and do not violate weak stochastic transitivity.

Weak stochastic transitivity is defined as:

If $P(A, B) \geq 0.5$ and $P(B, C) \geq 0.5$, then $P(A, C) \geq 0.5$.

Goldstone et al. chose weak stochastic transitivity as their measure of transitivity because it is "the easiest form of transitivity to satisfy, and hence, finding violations of this form of intransitivity is the most difficult and significant," (Goldstone, Medin, & Halberstadt, 1997, p. 240).

3.2 Study Design

Using a within-subject, two-way forced-choice design, the experiment presented all participants with three repetitions of the same 24 trios of audio stimuli for a total of 72 questions per participant. Each question included a reference stimulus and two test stimuli. Participants listened to all three stimuli and selected the test stimulus they judged more similar to the reference stimulus. The order of the questions and test stimuli per question was randomized. The experiment was conducted using the Qualtrics platform and a pair of Shure SRH-840A closed-back headphones in relatively quiet rooms, with no more than quiet speaking voices as ambient sound. A post-survey questionnaire asked about participants' ages, hearing impairment, handedness, years of experience playing music, and music listening habits.

3.3 Stimuli

A set of 8 reference stimuli and 21 test stimuli were synthesized using additive synthesis via the Analog instrument in Ableton Live 12. The stimuli were varied along three auditory dimensions shown by (Peeters, Giordano, Susini, Misdariis, & McAdams, 2011) to be relevant for timbre perception and to be relatively independent: spectral centroid (median), spectral variation (inter-quartile range), and attack slope. These attributes were measured using the Timbre Toolbox by (Peeters, Giordano, Susini, Misdariis, & McAdams, 2011) in Matlab. The stimuli were roughly loudness-equalized using the judgment of the experimenter. Each 440 Hz stimulus was 800 ms long followed by a 300 ms decay time.

Each *stimulus set* consisted of a reference stimulus and three test stimuli. The eight reference stimuli occupied the most extreme positions along all three auditory dimensions. The 21 test stimuli were linearly spaced at even intervals between the extremes along all three auditory dimensions. The stimulus sets were constructed so that each test stimulus was more similar to the reference than another along two auditory dimensions and less similar along the third dimension. There was therefore no ground truth similarity for any question. Questions were constructed using a Latin squares design, and every permutation of reference stimulus and three test stimuli was tested (see Appendix A). Participants answered a total of 72 questions consisting of three repetitions of the 24 stimulus permutations.

4. Analysis

Three analyses were carried out on the resulting data. 1) Participants' judgments were analyzed for itra-rater consistency across repetitions of identical questions. 2) The frequencies of the participants' answer patterns were counted and compared to the distribution of answer patterns predicted by both the null hypothesis and the alternative hypothesis using chi-squared goodness-of-fit tests. 3) Participants'

answer patterns were compared to the answer patterns expected if each of the three auditory dimensions dominated the similarity judgments.

4.1 Self-consistency of Participants

Each participant received each question three times. They could either provide the same answer for all three trials, or they could answer the question one way during one trial and the other way during the other two trials. I therefore counted the number of times each participant gave a consistent answer across all three repetitions and divided this by 24, the total number of three-way comparisons presented. The distribution of self-consistency of participants can be seen in Table 1.

Participant	Self-Consistency %
1	62.5 %
2	79.2 %
3	54.2 %
4	62.5 %
5	58.3 %
6	41.7 %
7	70.8 %
8	79.2 %
9	70.8 %
10	79.2 %

Table 1: Percent of 24 questions for which each participant offers consistent answers. Participants with lower self-consistency scores did not possess less experience with music listening or playing.

I also calculated the number of participants who gave self-consistent answers across all trials for each of the 24 questions. No question was answered with self-consistency by all participants, implying that all questions possessed some degree of ambiguity.

4.2 Transitivity of Similarity Judgments

The two hypothesized answer patterns and the observed answer patterns can be seen in Figure 4.

Hypothesized Distribution of Answer Patterns (N1)					
A>B	B>C	C>A	Overall Frequency	Prob. Dist.	% Dist.
Y	Y	Y	32	0.4	40.00%
Y	Y	N	6.67	0.08	8.33%
Y	N	Y	6.67	0.08	8.33%
Y	N	N	6.67	0.08	8.33%
N	Y	Y	6.67	0.08	8.33%
N	Y	N	6.67	0.08	8.33%
N	N	Y	6.67	0.08	8.33%
N	N	N	8	0.1	10.00%

a)

Null Hypothesis (N0)					
A>B	B>C	C>A	Overall Frequency	Prob. Dist.	% Dist.
Y	Y	Y	10	0.125	12.50%
Y	Y	N	10	0.125	12.50%
Y	N	Y	10	0.125	12.50%
Y	N	N	10	0.125	12.50%
N	Y	Y	10	0.125	12.50%
N	Y	N	10	0.125	12.50%
N	N	Y	10	0.125	12.50%
N	N	N	10	0.125	12.50%

b)

Observed Results					
A>B	B>C	C>A	Overall Frequency	Prob. Dist.	% Dist.
Y	Y	Y	3	0.0375	3.75%
Y	Y	N	5	0.0625	6.25%
Y	N	Y	18	0.225	22.50%
Y	N	N	5	0.0625	6.25%
N	Y	Y	21	0.2625	26.25%
N	Y	N	12	0.15	15.00%
N	N	Y	15	0.1875	18.75%
N	N	N	1	0.0125	1.25%

c)

Figure 4: a) Answer patterns expected if the hypothesis is correct. This distribution of answer patterns mirrors that observed by Goldstone et al. when performing frequency counts of similarity judgments of complex visual stimuli. b) Answer patterns expected if the null hypothesis, a random distribution across all answer patterns, was observed. c) Observed answer patterns. The observed answer patterns were shown to be significantly different from both the null hypothesis and the alternative hypothesis distributions using chi-squared goodness-of-fit tests.

The distribution of observed answer patterns is not significantly similar to the answer pattern expected if the hypothesis were correct, as established by a chi-squared goodness-of-fit test (p-value so close to zero as to be represented as zero in Google Sheets). The observed answer patterns do not violate stochastic weak intransitivity, invalidating the experimental hypothesis. The observed answer patterns are also not significantly similar to that implied by the null hypothesis, with a chi-squared p-value = 1.6e-6. The results are, further, not significantly similar to a distribution featuring all-transitive answer patterns, with a p-value = 1.5e-6. None of the proposed hypotheses fits the observed patterns of answers.

4.3 Dominant Audio Attributes

For each of the eight stimulus sets, the collected similarity judgments of all participants across trials was examined to see if any particular auditory attribute dominated the similarity judgments for that particular stimulus set. See Appendix B for tables showing all answer patterns separated by stimulus set.

Spectral centroid (median) correlates with the majority of similarity judgments in 6.25% of cases. Spectral variation (inter-quartile range) correlates with the majority of similarity judgments in 0.13% of cases. Attack slope correlates with the majority of similarity judgments in 28.75% of cases. In 55% of cases, none of the three auditory attributes correlates with the similarity judgments. In a few of the stimulus sets, a large majority of participants (8/10) provided the same answer pattern that did not correspond to intentional variation of any of the three auditory dimensions. There may be unaccounted-for auditory or associative attributes influencing similarity judgments in these cases.

5. Discussion

Neither the hypothesis nor the null hypothesis is supported by the observed timbre similarity judgments. The similarity judgments made by the participants may reflect perceptual processes unaccounted for in the experiment design. No single auditory dimension dominates the similarity judgments of the participants, although attack slope seems more relevant for timbre similarity judgments than spectral centroid (median) or spectral variation (inter-quartile range). Further refinement of the study will be needed to understand the processes at work.

5.1 Future Work

There are many opportunities for improvement of the current study. The Timbre Toolbox (Peeters, Giordano, Susini, Misdariis, & McAdams, 2011) can extract 54 different auditory attributes, and

these could all be extracted and compared with the observed similarity judgments to determine if a different auditory attribute may be correlated to some of the judgments. This could be done on a participant-by-participant basis as well to understand if there are meaningful groups of timbre listeners among the participants. After feedback from a few participants who heard the stimuli as multiple layered sounds rather than a single tone, the stimuli should be re-synthesized using a non-additive approach to achieve more cohesive sounds. Canonical intra- and inter-rater reliability metrics should be employed to eliminate particularly non-self-consistent participants, although the wide range of self-consistency among participants may indicate interesting differences in timbre perception. In the original method, Goldstone et al. engaged each participant in an extensive stimulus customization process to ensure that each stimulus was spaced equally along each perceptual dimension according to the perception of that individual. A similar process may be necessary for these timbral stimuli. It's possible, and maybe likely, that linear spacing of the stimuli along each of the three auditory dimensions does not achieve perceptually equal spacing.

6. Conclusion

Although more refinement of the experiment is needed to draw conclusions about the transitivity of timbre similarity judgments, the pilot study provided valuable feedback for improvement of the study design. The wide variety of verbal responses to the stimuli were surprising, with some participants describing aspect of the sounds as "bells" or "wind" and others describing the sounds with precise acoustic terminology. Participants asked if they should ignore quite a few different aspects of the sounds, including the attacks, higher partials, and noisy aspects, and several participants expressed that they had deliberately tried to ignore these things throughout the experiment, assuming them to be irrelevant or maskers. After completing the pilot study, I feel I know no more about timbre than when I started, but future iterations of the experiment are needed to offer conclusive results.

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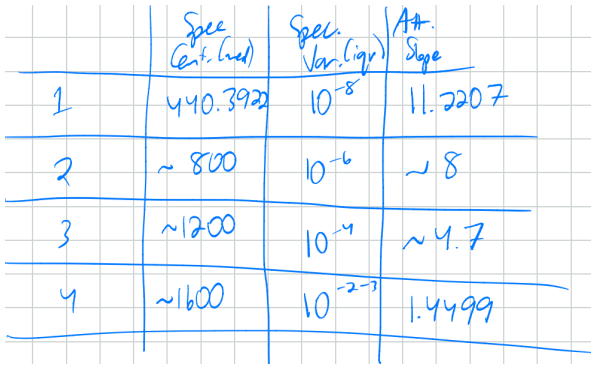
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Appendix A: Stimulus Design

This section includes two screenshots of hand-drawn charts of the spectral centroid, spectral variation, and attack slope values of the stimuli as well as the answer patterns expected if each of those auditory attributes dominated the responses for each stimulus set. I didn't re-format these as nice-looking tables since I know I'm going to be re-synthesizing my stimuli differently next time.





Appendix B: Dominance of Auditory Attributes by Stimulus Set

Ref: 111	A>B	B>C	C>A	Part. 0	Part. 1	Part. 2	Part. 3	Part. 4	Part. 5	Part. 6	Part. 7	Part. 8	Part. 9	Frequency			Dominant Audio Attribute
	Y	Y	Y							1				1			
	Y	Y	N											0	A/B/C	spectral variation (iqr)	
	Y	N	Y			1								1	C/A/B	attack slope	
	Y	N	N	1					1					2	A/C/B		
	N	Y	Y								1			1	B/C/A	spectral centroid (med)	
	N	Y	N		1			1				1	1	4	* B/A/C		
	N	N	Y				1							1	C/B/A		
	N	N	N											0			

Ref: 114	A>B	B>C	C>A	Part. 0	Part. 1	Part. 2	Part. 3	Part. 4	Part. 5	Part. 6	Part. 7	Part. 8	Part. 9	Frequency			
	Y	Y	Y											0			
	Y	Y	N							1				1	A/B/C	spectral centroid (med)	
	Y	N	Y		1	1		1			1	1	1	6	* C/A/B	spectral variation (iqr)	
	Y	N	N											0	A/C/B		
	N	Y	Y											0	B/C/A	attack slope	
	N	Y	N											0	B/A/C		
	N	N	Y	1			1		1					3	C/B/A		
	N	N	N											0			

Ref: 441	A>B	B>C	C>A	Part. 0	Part. 1	Part. 2	Part. 3	Part. 4	Part. 5	Part. 6	Part. 7	Part. 8	Part. 9	Frequency			
	Y	Y	Y						1			1		2			
	Y	Y	N							1				1	A/B/C	spectral centroid (med)	
	Y	N	Y			1								1	C/A/B	spectral variation (iqr)	
	Y	N	N											0	A/C/B		
	N	Y	Y					1						1	B/C/A	attack slope	
	N	Y	N		1		1				1			3	* B/A/C		
	N	N	Y										1	1	C/B/A		
	N	N	N	1										1			

Ref: 141	A>B	B>C	C>A	Part. 0	Part. 1	Part. 2	Part. 3	Part. 4	Part. 5	Part. 6	Part. 7	Part. 8	Part. 9	Frequency			
	Y	Y	Y											0			
	Y	Y	N						1					1	A/B/C	spectral centroid (med)	
	Y	N	Y											0	C/A/B	spectral variation (iqr)	
	Y	N	N											0	A/C/B		
	N	Y	Y		1	1	1			1	1			5	* B/C/A	attack slope	
	N	Y	N					1				1	1	3	B/A/C		
	N	N	Y	1										1	C/B/A		
	N	N	N											0			

Ref: 144	A>B	B>C	C>A	Part. 0	Part. 1	Part. 2	Part. 3	Part. 4	Part. 5	Part. 6	Part. 7	Part. 8	Part. 9	Frequency			
	Y	Y	Y											0			
	Y	Y	N											0	A/B/C	spectral centroid (med)	
	Y	N	Y											0	C/A/B	spectral variation (iqr)	
	Y	N	N						1					1	A/C/B		
	N	Y	Y		1	1	1	1		1	1	1	1	8	* B/C/A	attack slope	
	N	Y	N	1										1	B/A/C		
	N	N	Y											0	C/B/A		
	N	N	N											0			

Ref: 411	A>B	B>C	C>A	Part. 0	Part. 1	Part. 2	Part. 3	Part. 4	Part. 5	Part. 6	Part. 7	Part. 8	Part. 9	Frequency			
	Y	Y	Y											0			
	Y	Y	N											0	A/B/C	spectral centroid (med)	
	Y	N	Y	1	1	1	1	1			1	1	1	8	* C/A/B	*this doesn't fit any of the three attributes*	
	Y	N	N											0	A/C/B	spectral variation (iqr)	
	N	Y	Y						1					1	B/C/A		
	N	Y	N											0	B/A/C	spectral centroid (med)	
	N	N	Y							1				1	C/B/A	attack slope	
	N	N	N											0			

Ref: 414	A>B	B>C	C>A	Part. 0	Part. 1	Part. 2	Part. 3	Part. 4	Part. 5	Part. 6	Part. 7	Part. 8	Part. 9	Frequency			
	Y	Y	Y											0			
	Y	Y	N						1					1	A/B/C	spectral centroid (med)	
	Y	N	Y											0	C/A/B	spectral variation (iqr)	
	Y	N	N											0	A/C/B		
	N	Y	Y			1	1	1			1	1		5	* B/C/A	attack slope	
	N	Y	N							1				1	B/A/C		
	N	N	Y	1	1								1	3	C/B/A		
	N	N	N											0			

Ref: 444	A>B	B>C	C>A	Part. 0	Part. 1	Part. 2	Part. 3	Part. 4	Part. 5	Part. 6	Part. 7	Part. 8	Part. 9	Frequency			
	Y	Y	Y											0			
	Y	Y	N			1								1	A/B/C	spectral variation (iqr)	
	Y	N	Y									1	1	2	C/A/B	attack slope	
	Y	N	N	1				1						2	A/C/B		
	N	Y	Y											0	B/C/A	spectral centroid (med)	
	N	Y	N											0	B/A/C		
	N	N	Y		1		1		1	1	1			5	* C/B/A	*this doesn't fit any of the three attributes*	
	N	N	N											0			

Observed Results

	A>B	B>C	C>A	Overall Frequency	Prob. Dist.	% Dist.
	Y	Y	Y	3	0.0375	3.75%
	Y	Y	N	5	0.0625	6.25%
	Y	N	Y	18	0.225	22.50%
	Y	N	N	5	0.0625	6.25%
	N	Y	Y	21	0.2625	26.25%
	N	Y	N	12	0.15	15.00%
	N	N	Y	15	0.1875	18.75%
	N	N	N	1	0.0125	1.25%
				80		

NULL HYP. N0

	A>B	B>C	C>A	Overall Frequency	Prob. Dist.	% Dist.	Chi-Square Test	
	Y	Y	Y	0.33	0.00412534	0.41%		
	Y	Y	N	13.00	0.16	16.25%	0.0000015164546	chi square p-value
	Y	N	Y	13.33	0.17	16.67%		significant difference
	Y	N	N	13.33	0.17	16.67%		
	N	Y	Y	13.33	0.17	16.67%		
	N	Y	N	13.33	0.17	16.67%		
	N	N	Y	13.33	0.17	16.67%		
	N	N	N	13.00	0.16	16.25%		
	N	N	N	0.33	0.00412534	0.41%		

Null Hypothesis (N0)

	A>B	B>C	C>A	Overall Frequency	Prob. Dist.	% Dist.	Chi-Square Test	
	Y	Y	Y	10	0.125	12.50%		
	Y	Y	N	10	0.125	12.50%	0.0000016393599	chi square p-value
	Y	N	Y	10	0.125	12.50%		significant difference
	Y	N	N	10	0.125	12.50%		
	N	Y	Y	10	0.125	12.50%		
	N	Y	N	10	0.125	12.50%		
	N	N	Y	10	0.125	12.50%		
	N	N	N	10	0.125	12.50%		

Hypothesized Distribution of Answer Patterns (N1)

	A>B	B>C	C>A	Overall Frequency	Prob. Dist.	% Dist.	Chi-Square Test	
	Y	Y	Y	32	0.4	40.00%		
	Y	Y	N	6.67	0.08	8.33%	0	chi square p-value
	Y	N	Y	6.67	0.08	8.33%		significant difference
	Y	N	N	6.67	0.08	8.33%		
	N	Y	Y	6.67	0.08	8.33%		
	N	Y	N	6.67	0.08	8.33%		
	N	N	Y	6.67	0.08	8.33%		
	N	N	N	8	0.1	10.00%		

Spectral Centroid (med) Total Count (/80)

5	0.0625	6.25%

Spectral Variation (iqr) Total Count (/80)

8	0.10	0.13%

Attack Slope Total Count (/80)

23	0.29	28.75%

None Of The Above

44	0.55	55.00%